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Subject: Application for the M2 Internship – Sol-Gel and Flame Spray Pyrolysis Approach for Layered Oxide Cathode Materials

Dear Dr. Leconte, Dr. Yagoubi, Dr. Pongilat, Dr. Bertrand, and Dr. Foy,

I am writing to apply for the M2 internship on combining sol-gel precursor chemistry with Flame Spray Pyrolysis for the synthesis of layered oxide cathode materials for Li-ion batteries, at CEA Paris-Saclay (LEEL/LAPA). I am currently in my final year of the Industrial Chemical engineering program at INSAT (University of Carthage), specializing in Process and Environment Engineering, and I am seeking a 6-month end-of-studies internship from February 2027.

This project sits directly at the intersection of my two main areas of experience: battery materials and sol-gel/nanoparticle synthesis. My last internship at SEGULA Technologies focuses on electrochemical modeling of a Li-ion cell (NMC811/Graphite-SiOx) using the Single Particle Model in MATLAB, solved via finite-volume discretization with Crank-Nicolson time integration and validated against COMSOL and experimental data, which gives me a strong working knowledge of the electrochemical behavior your cathode materials are designed to improve. On the synthesis side, I have hands-on experience with sol-gel-type nanomaterial synthesis and structural characterization (XRD, SEM/TEM), including a core-shell MOF/polymer system built via controlled precipitation and surface functionalization. I am also familiar with electrochemical characterization techniques (cyclic voltammetry, polarography) from a separate project. Flame Spray Pyrolysis and half-cell galvanostatic cycling are new to me specifically, but my track record across synthetic, structural, and electrochemical projects shows I adapt quickly to new techniques and lab environments.

This project particularly appeals to me because it combines precursor chemistry, nanomaterial synthesis, and battery electrochemistry and this is the exact combination I have been building toward through my academic and internship experience, while addressing a concrete scale-up challenge for next-generation cathode materials.

I would welcome the opportunity to discuss my application further and thank you for considering my candidacy.

Sincerely,
Nermine Moumen